6.1 Discrete Probability Distribution

A reference sheet for the distributions in Chapter 6 and 7.

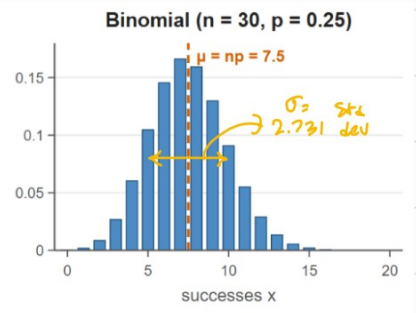
Throughout p = probability of success, $q = 1-p$ = probability of failure.

1. Binomial Distribution (Discrete)

X = number of successes in n independent trials.

$$P(X=x) = {}_nC_x\, p^x q^{n-x} \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

Mean	Standard Deviation
$\mu = np$	$\sigma = \sqrt{npq}$



$$\mu = np = 30 \times 0.25 = 7.5 \qquad \sigma = \sqrt{npq} = \sqrt{30 \times 0.25 \times 0.75} = 2.371$$

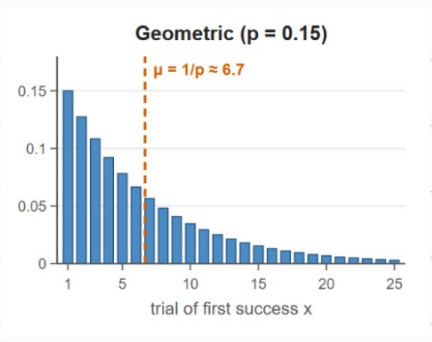
2. Geometric Distribution (Discrete)

X = The trial on which first success happens.

$$P(X=n) = q^{n-1}p$$

Why: for the first success to be on trial 4, trial 1-3 must fail (q.q.q.) and trial 4 must succeed (p), so $P(X=4) = q^3p$

Probability	Formula
$P(X>n)$	q^n
$P(X \geq n)$	q^{n-1}
$P(X \leq n)$	$1-q^n$
$P(X<n)$	$1-q^{n-1}$



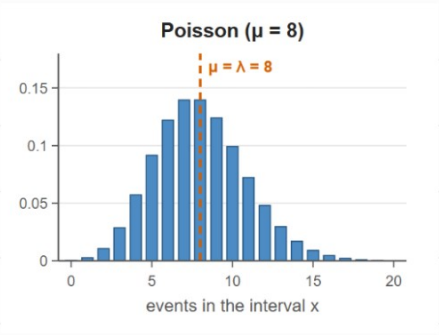
Mean	$\mu = \frac{1}{p}$
Variance	$\sigma^2 = \frac{1}{p}\left(\frac{1}{p}-1\right) = \mu(\mu-1) = \frac{q}{p^2}$
Standard Deviation	$\sigma = \frac{\sqrt{1-p}}{p}$

3. Poisson Distribution (Discrete)

X = number of events in a fixed interval, where the average number of events is $\mu = \lambda$

$$P(X=n) = \frac{\mu^n e^{-\mu}}{n!} = \frac{\lambda^n e^{-\lambda}}{n!}$$

Probability	Formula
$P(X \le n)$	$e^{-\mu} \sum_{x=0}^{n} \frac{\mu^x}{x!}$
$P(X > n$	$1 - e^{-\mu} \sum_{x=0}^{n} \frac{\mu^x}{x!}$



Mean	$\mu = \lambda$ (= np when approximating a binomial)
Variance	$\sigma^2 = \lambda$
Standard Deviation	$\sigma = \sqrt{\lambda}$

4. Uniform Distribution (Continuous)

X is equally likely to be anywhere between a and b. The graph is rectangle of width $b-a$, and its total area must be 1:

$$\text{Area} = \text{Base} \times \text{Height} = (b-a) \cdot f(x) = 1 \rightarrow f(x) = \frac{1}{b-a}$$

Probability = Partial Width / Total Width:

$$P(c \le X \le d) = \frac{d-c}{b-a}$$

For example, $P(a \le X \le c) = \frac{c-a}{b-a}$ and $P(c \le X \le b) = \frac{b-c}{b-a}$

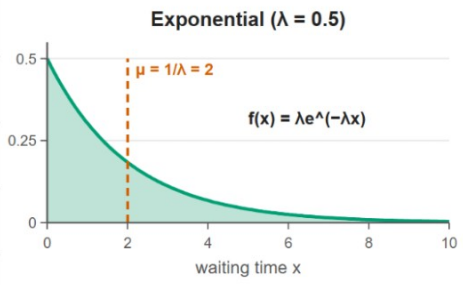
Mean	Standard Deviation
$\mu = \frac{a+b}{2}$	$\sigma = \frac{b-a}{\sqrt{12}}$

5. Exponential Distribution (Continuous)

Waiting times. λ = rate parameter, and $\mu = 1/\lambda$

$$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0$$

Probability	Formula	From Calculus
$P(X \le n)$ (left area)	$1 - e^{-\lambda n}$	$\int_0^n \lambda e^{-\lambda x} dx$
$P(X \ge n)$ (right area)	$e^{-\lambda n}$	$\int_n^\infty \lambda e^{-\lambda x} dx$



Mean	Variance	Rate
$\mu = \frac{1}{\lambda}$	$\sigma^2 = \frac{1}{\lambda^2}$	$\lambda = \frac{1}{\mu}$

* In general, a probability for a continuous distribution is the area under $f(x)$: $P(a \le X \le b) = \int_a^b f(x)\,dx$

- Where each one is used

Distribution	Typical Question
Binomial	"Exactly / at least x successes in n trials"
Geometric	"First success on trial n"
Poisson	"Number of events in a time or space interval"
Exponential	"Waiting time until the next event"

6.2 Finding the Probability of a Binomial Distribution

1. Formula

$$P(X=x) = {}_nC_x\, p^x q^{n-x}, \quad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

Mean: $\mu = np$, Standard Deviation: $\sigma = \sqrt{npq}$

Symbol	Meaning
n	Number of Trials
X	Number of Successes you want (exactly x)
p	Probability of Success on One Trial
$q = 1-p$	Probability of Failure on One Trial

Conditions :-
- A fixed number of trials
- The trials are independent
- And p is always the same on every trial.

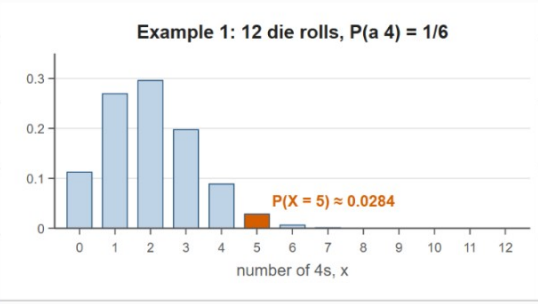
2. Example 1 - rolling a die.

A die is rolled 12 times. What is the probability of getting a 4 exactly five times?

- $n = 12$, $x = 5$, $p = \frac{1}{6}$

$$P(X=x) = {}_nC_x\, p^x q^{n-x} \qquad {}_nC_x = \frac{n!}{(n-x)!\,x!}$$

$$P(X=5) = {}_{12}C_5 \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^{12-5} = \frac{12!}{(12-5)!\,5!}\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^7 = 792 \times \left(\frac{1}{6}\right)^5 \times \left(\frac{5}{6}\right)^7 = 0.0284 = 2.84\%$$



2. Example 2 - Guessing on a multiple-choice test

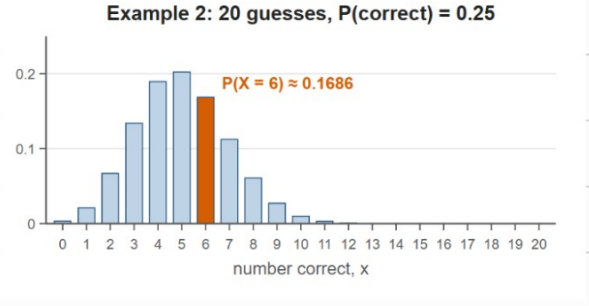
A test has 20 questions, each with 4 choice (A-D) and one correct answer. A student guesses on every question.

What is the probability of exactly 6 correct?

- $n = 20$, $x = 6$, $p = \frac{1}{4}$, $q = \frac{3}{4}$

$$P(X=6) = {}_{20}C_6\, p^6 q^{14} = \frac{20!}{(20-6)!\,6!} \times \left(\frac{1}{4}\right)^6 \times \left(\frac{3}{4}\right)^{14} = 0.1686 = 16.86\%$$

(With 5 answer choice instead, $p = 0.20$ and $q = 0.80$)



3. Example 3 - Students Taking Algebra.

25% of Students at a School take algebra. 30 Students are chosen at random.

(a) P(exactly 7 take algebra)

$p = \frac{1}{4}, q = 1-p = \frac{3}{4}, x = 7, n = 30$

$$P(X=7) = {}_{30}C_7\, p^7 q^{30-7} = \frac{30!}{(30-7)!\,7!}\left(\frac{1}{4}\right)^7\left(\frac{3}{4}\right)^{30-7} = 0.1662 = 16.62\%$$

(b) P(fewer than 5 take algebra) so $X < 5$, meaning $X = 0, 1, 2, 3, 4$, (5 not included)

x	$P(X = x) = {}_{30}C_x(0.25)^x(0.75)^{30-x}$
0	0.000179
1	0.001786
2	0.008631
3	0.026853
4	0.060420
Sum	**0.09787 ≈ 9.79%**

Correction: the video adds only $x = 1$ to 4 and gets 9.77%. "Fewer than 5" also includes $x = 0$ (no algebra students at all), which adds 0.00018, so the correct total is about **9.79%**. The difference is tiny here, but leaving out $x = 0$ is a common mistake.

For large n, adding many terms gets tedious. A normal approximation, or the complement rule with a calculator's binomial CDF, is faster.

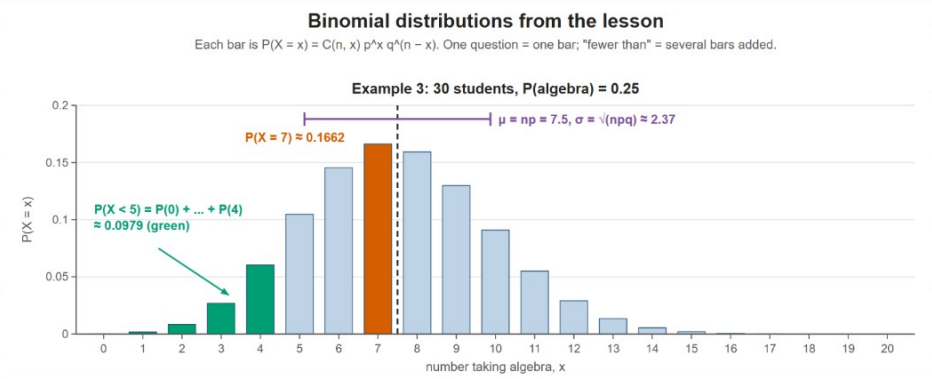
$$\mu = np = 30 \times \frac{1}{4} = 7.5$$

$$P(X \le x) = e^{-\mu}\sum_{x=0}^{n}\frac{\mu^x}{x!} = e^{-7.5}\sum_{x=0}^{30}\frac{7.5}{x!} = 0.11275 = 11.275\%$$

(c) Mean and Standard deviation

$$\mu = np = 30(0.25) = 7.5$$

$$\sigma = \sqrt{npq} = \sqrt{30(0.25)(0.75)} = \sqrt{5.625} = 2.37$$



4. Key Takeaways

- Identify n, x, p and q first, then substitute ${}_nC_x\, p^x q^{n-x}$
- "Exactly x" → One term "Fewer than / at most / at least" → add Several terms and remember $x=0$ when it applies.
- $\mu = np$ and $\sigma = \sqrt{npq}$

6.3 The Geometric Distribution

1. The Idea

A geometric distribution answer: On which trial does the first success happen?

If the first successfull is on trial x, then the first $x-1$ trials failed and trial x succeeded

$P(X=x) = q^{x-1}p, \quad q=1-p$

2. All the formulas

Question	Formula
First Success on trial x	$P(X=x) = q^{x-1}p$
First Success by trial x (within the first x)	$P(X\leq x) = 1-q^x$
First Success before trial x	$P(X<x) = 1-q^{x-1}$
First Success on trial x or later	$P(X>x) = q^x$
Mean (expected trial of the first Success)	$\mu = \frac{1}{p}$
Variance	$\sigma^2 = \frac{1}{p}\left(\frac{1}{p}-1\right) = \frac{1-p}{p^2}$
Standard Deviation	$\sigma = \sqrt{\frac{1-p}{p^2}} = \frac{\sqrt{1-p}}{p}$

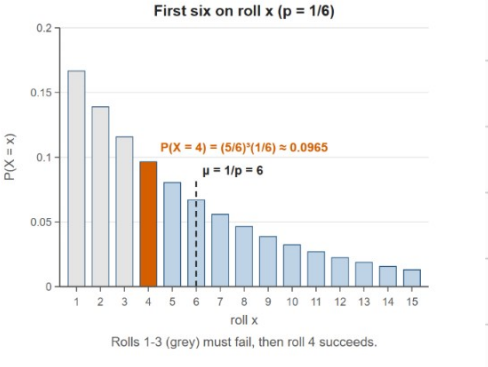
Why $P(X>x) = q^x$: "no success in the first x trials" means x failure in a row. Then $P(X\leq x) = 1-q^x$ is its complement.

Tip for $<$ and $\geq$: Shift by one so you can use the formula above $P(X<7) = P(X\leq 6)$ and $P(X\geq 18) = P(X>17)$

3. Example 1 - First Six on the Fourth Roll

$p = \frac{1}{6}$ (rolling a 6), $q = \frac{5}{6}$

$P(X=4) = q^{4-1}p = \left(\frac{5}{6}\right)^3 \times \left(\frac{1}{6}\right) = 9.645\%$



4. Example 2 - First blue car is the 7th Car

15% of Cars are blue: $p = 0.15$, $q = 0.85$

$P(X=7) = q^{7-1}p = 0.85^6 \times 0.15 = 5.657\%$

5. Example 3 - The 10th Person You Meet is The First Teacher

4% of Town's Resident are teacher: $p = 0.04$ $q = 0.96$

- $P(X=10) = q^{10-1}p = 0.96^9 \times 0.04 = 2.77\%$
- Mean $\mu = \frac{1}{p} = \frac{1}{0.04} = 25$ • Variance $\sigma^2 = \frac{1-p}{p^2} = \frac{0.96}{0.04^2} = 600$ • Standard Deviation $\sigma = \frac{\sqrt{1-p}}{p} = \frac{\sqrt{0.96}}{0.04} = 24.494$

6. Example 4 - Defective Tires ($p = 0.02$, $q = 0.98$)

2% of tires are defective, and a sample of tires is tested one at a time.

(a) The 8th tire is the first defective one

$$P(X = 8) = q^{8-1}p = 0.98^7 \times 0.02 = 0.0173 = 1.73\%$$

(b) The first defect is found within the first 5 tires.

$$P(X \leq 5) = P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

With the cumulative formula, which is much faster

$$P(X \leq 5) = 1 - q^x = 1 - 0.98^5 \approx 0.0961 = 9.6\%$$

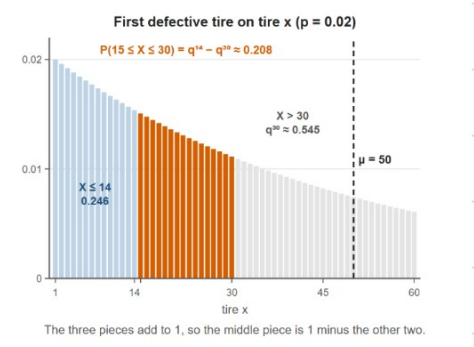
(c) The first defect is found within the first 10

$$P(X \leq 10) = 1 - (0.98)^{10} = 18.3\%$$

(d) How many tires do you expect to test to find the first defect

$$\mu = \frac{1}{p} = \frac{1}{0.02} = 50 \text{ tires.}$$

7. Example 5 - More Cumulative Questions (Same Tires)



(a) First defect before the 7th Tire

$$P(X < 7) = P(X \leq 6) = 1 - (0.98)^6 = 0.1142 = 11.42\%$$

(b) First defect after the 12th Tire

$$P(X > 12) = q^{12} = 0.98^{12} = 0.7847 = 78.47\%$$

(c) First defect on the 18th tire or later

$$P(X \geq 18) = q^{18-1} = 0.98^{18-1} = 70.93\%$$

(d) First defect between tire 15 and tire 30 (Inclusive)

Split the whole range (total probability 1) into three pieces:

$A = X \leq 14$, $B = 15 \leq X \leq 30$, $C = X > 30$

$$P(15 \leq X \leq 30) = 1 - P(X \leq 14) - P(X > 30)$$

$$= 1 - [1 - (0.98)^{14}] - (0.98)^{30} = (0.98)^{14} - (0.98)^{30}$$

$$= 0.7536 - 0.5455 = 0.2082 = 20.8\%$$

(e) First defect after tire 20 but no later than tire 45 ($21 \leq X \leq 45$)

$$P(21 \leq X \leq 45) = 1 - P(X \leq 20) - P(X > 45) = (0.98)^{20} - (0.98)^{45} = 26.5\%$$

8. Key Takeaways

- Geometric = Waiting for the first Success: $P(X = x) = q^{x-1}p$
- Cumulative: $P(X \leq x) = 1 - q^x$ and $P(X > x) = q^x$ Convert $<$ and $\geq$ by shifting by 1
- $\mu = 1/p$ is the expected number of trials until the first Success.

6.4 The Poisson Distribution

1. The Idea

The **Poisson distribution** gives the probability of **exactly x events** in a **fixes intervals** (of time, area, ...) when the events happen at a known **average rate** μ (also written λ).

$$P(X=x) = \frac{\mu^x e^{-\mu}}{x!}$$

- $e = 2.7182$
- $x! = x(x-1)\cdots 2.1$ with $0! = 1$

The mean must match the interval

If the interval changes, scale the mean in proportion $\frac{\mu_2}{\mu_1} = \frac{t_2}{t_1}$

2. Example 1 - Customer per day

A business averages **12 customer per day** what is the probability of exactly **8** customer in one day?

$\mu = 12$, $x = 8$

$$P(X=8) = \frac{\mu^x e^{-\mu}}{x!} = \frac{12^8 e^{-12}}{8!} = 6.55\%$$

3. Example 2 - Text Messages.

A student receive on average **7 text per 2 hours**

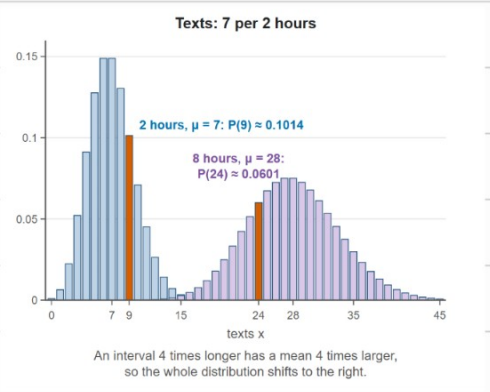
(a) Exactly **9 texts in 2 hours**. The interval is the same, so $\mu = 7$:

$$P(X=9) = \frac{\mu^9 - e^{-\mu}}{x!} = \frac{7^9 \cdot 2.713^{-7}}{9!} = 10.14\%$$

(b) Exactly **24 texts in 8 hours**. The interval is 4 times longer, so the mean is 4 Times larger:

Calculate the mean $\frac{\mu_2}{\mu_1} = \frac{t_2}{t_1} \rightarrow \mu_2 = \frac{t_2}{t_1} \times \mu_1 = \frac{8}{2} \times 7 = 28$

$$P(X=24) = \frac{\mu_2^x e^{-\mu}}{x!} = \frac{28^{24} \cdot e^{-28}}{24!} = 6.009\%$$



An interval 4 times longer has a mean 4 times larger, so the whole distribution shifts to the right.

4. Example 3 – Phone Calls ($\mu = 8$ per hour)

(a) Exactly 7 Calls in One Hour

$$P(X=7) = \frac{\mu^x e^{-\mu}}{x!} = \frac{8^7 e^{-8}}{7!} = 13.96\%$$

(b) At Most 5 Calls ($X \leq$, so $X = 0, 1, \dots, 5$)

Factor Out $e^{-\mu}$, Since only $\mu^x/x!$ Changes from term to term:

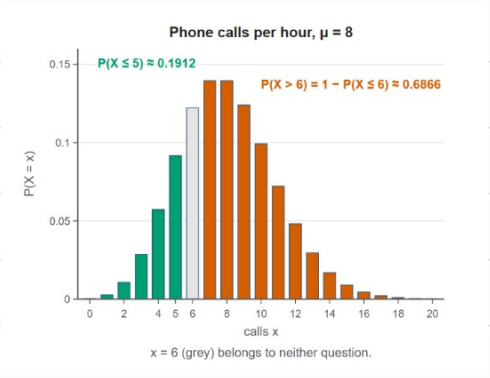
$$P(X \leq 5) = e^{-\mu} \sum_{x=0}^{5} \frac{\mu^x}{x!} = e^{-8} \sum_{x=0}^{5} \frac{8^x}{x!} = e^{-8}\left(\frac{8^0}{0!} + \frac{8^1}{1!} + \frac{8^2}{2!} + \frac{8^3}{3!} + \frac{8^4}{4!} + \frac{8^5}{5!}\right) = 19.12\%$$

(c) More Than 6 Calls ($X > 6$, which runs to Infinity)

Use the Complement, Since the total probability is 1

$$P(X>6) = 1 - P(X \leq 6) = 1 - e^{-8} \sum_{x=0}^{6} \frac{8^x}{x!} = 1 - e^{-8}(934.16) \approx 1 - e^{-8}(934.16) = 1 - 0.3134 = 0.6866 = 68.7\%$$

This is likely, which makes sense: the average is 8 calls, so more than 6 is the usual case



5. Key Takeaways

- Poisson: $P(X=x) = \frac{\mu^x e^{-\mu}}{x!}$ Counts events in a fixed interval.
- Adjust μ whenever the interval in the question differs from the one in the given rate.
- "At most k" → add $x=0$ to k "more than k" → $1 - P(X \leq k)$
- Mean = Variance = μ and the Standard deviation is $\sqrt{\mu}$

6.5 Continuous Vs Discrete Data

1. Definitions

	Continuous data	Discrete data
Values	**Any** value in an interval	Only **specific** values (often whole numbers)
How it's obtained	**Measured**	**Counted**
Values between 1 and 10	1, 1.3, 2.46, 5.389, ...	1, 2, 3, ..., 10
Example	Temperature: 83.6 °F, 99.46 °F	Rolling a die: 1-6 only (never 2.37)

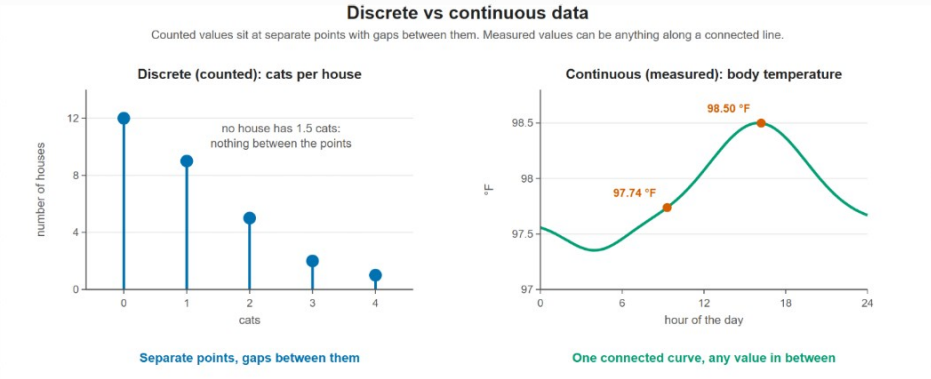
2. Practice - Continuous (C) or Discrete (D)?

#	Quantity	Answer	Reason
1	Speed of a car	**C**	Measured: 25.6 mph, 42.86 mph, ...
2	Number of cats in a neighborhood	**D**	Counted; no half cats
3	Your body weight	**C**	Measured: 160 lb, 173.4 lb, ...
4	Number of doctors in a town	**D**	Counted
5	Number of books on a shelf	**D**	Counted
6	Height of a student	**C**	Measured: 5.84 ft, 6.3 ft, ...
7	Number of students in a classroom	**D**	Counted
8	Your age **rounded** to the nearest whole number	**D**	Only whole numbers: 16, 19, 22, 24
9	Your **exact** age	**C**	Measured: 16.2, 19.3, 21.76, 23.958 years

Items 8 and 9 show that **rounding** a continuous quantity turns it into discrete data.

3. Recognizing them on a graph

- Continuous: One connected line or curve with no jumps or graph.
- Discrete: Separate points, or bars / step that don't connect into a smooth line.



Counted data (left) can only sit at whole numbers, so the graph is separate points. Measured data (right) can take any value, so the graph is one connected curve.

4. Key Takeaways

- Measured → Continuous (any value in range). Counted → Discrete (specific value only)
- Rounding turns continuous data into discrete data
- Graph
  - A smooth connected line means continuous.
  - Isolated points mean discrete.
- This distinction decides which distribution apply
  - binomial, geometric and poisson are discrete.
  - Uniform, exponential and normal are continuous.